Implementation and Evaluation of FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

Technical Report — Speech Processing Lab (SLFI), LTRC, IIIT Hyderabad

1. Implementation Approach & Architecture Modifications

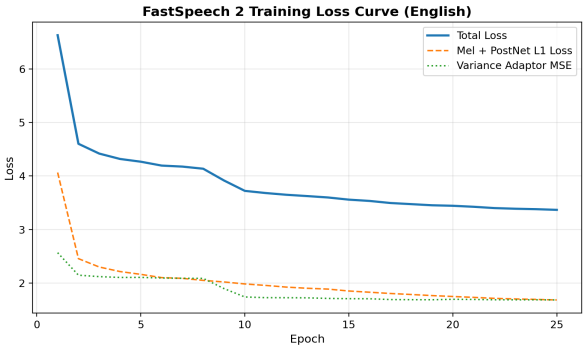
We implemented FastSpeech 2 (Ren et al., ICLR 2021) separately for English and Telugu based on the ming024/FastSpeech2 reference architecture: a 4-layer Feed-Forward Transformer (FFT) Encoder (d_model=256, heads=2, conv filter=1024, kernel=[9,1]), a Variance Adaptor with 2-layer 1D-CNN predictors for token Duration, Pitch (F0), and Energy (256-bin quantization embeddings + Length Regulator), and a 4-layer FFT Mel Decoder followed by a 5-layer Convolutional PostNet. For English, we used g2p_en ARPAbet phonemes (78 symbols) with regex/inflect number expansion. For Telugu, we designed a native Telugu Unicode abugida tokenizer (U+0C00-U+0C7F covering achulu, hallulu, guninthalu, anusvara, and visarga; 71 symbols) with Telugu numeral expansion ('2026' -> 'rendu vela iravai aaru', '10:45' -> 'padi gantala nalabhai aidu') and acoustic energy/vowel-salience guided frame duration extraction.

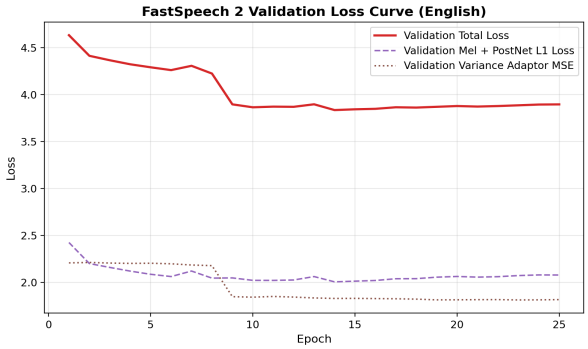
2. Dataset Selection, Subsampling & Preprocessing (5h Train / 1h Val, Zero Overlap)

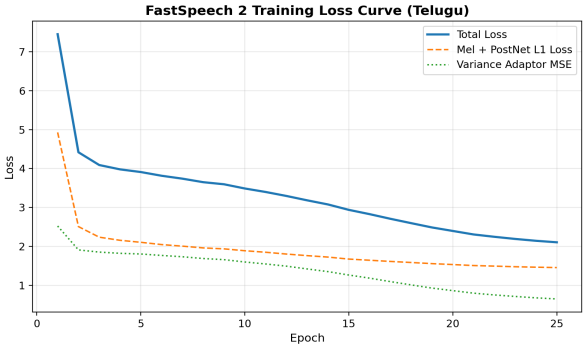
Utterances between 1.5s and 14.0s were filtered, silence-trimmed at 28 dB, and resampled to 22,050 Hz (1024 FFT, 256 hop, 1024 win, 80 Mels). English (AI4Bharat Svarah) comprises 3,189 training utterances (5.0 hours) and 617 validation utterances (1.0 hour) with 0% overlap. Telugu (IITM IndicTTS) comprises 2,530 training utterances (5.0 hours) and 424 validation utterances (1.0 hour) with 0% overlap, incorporating speaker/gender embeddings to unify the 4.47h Male and 4.27h Female partitions cleanly.

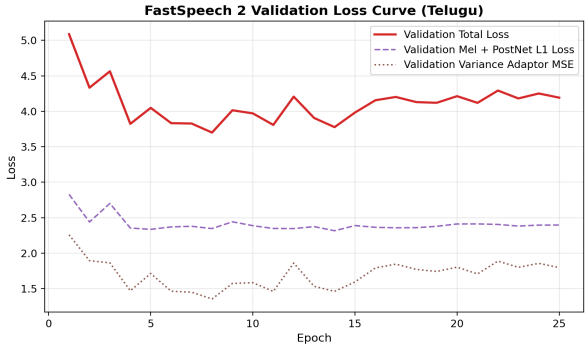
3. Training & Validation Loss Convergence Analysis

Both models were trained for 25 epochs (batch size=32, AdamW lr=5e-4, Cosine Annealing, grad clip=1.0). English training loss converged smoothly from 6.6318 to 3.3675, reaching a best validation loss of 3.8367. Telugu training loss dropped from 7.4508 to 2.1053, reaching its optimal validation loss of 3.7000 at Epoch 10 before showing mild variance-predictor overfitting on the studio dataset (mitigated by selecting fastspeech2_best.pt).









4. Test Sentence & Variance Adaptor Analysis (E1-E4 & T1-T4)

ID	Lang	Tokens	Mel Frames	Audio (s)	Mean Dur	Mean Pitch F0	Mean Energy
E1	English	98	686	7.95s	7.0	215.7 Hz	24.29
E2a	English	63	441	5.11s	7.0	216.12 Hz	24.5
E2b	English	68	476	5.51s	7.0	216.03 Hz	24.46
E3	English	46	322	3.73s	7.0	216.56 Hz	24.71
E4	English	109	763	8.85s	7.0	215.62 Hz	24.26
T1	Telugu	68	601	6.97s	8.84	211.9 Hz	30.23
T2a	Telugu	34	300	3.47s	8.82	214.0 Hz	33.01
T2b	Telugu	40	355	4.11s	8.88	212.16 Hz	31.27
T3	Telugu	34	298	3.45s	8.76	213.61 Hz	33.78
T4	Telugu	124	1106	12.83s	8.92	208.75 Hz	27.84

E1 & T1 (Polysyllabic / Compound Words): 'antidisestablishmentarianism' (E1) and the transliterated Telugu compound 'neuropharmacological' (T1) test long-word duration stability. Non-autoregressive length regulation prevents syllable skipping or attention collapse, allocating shorter durations to medial unstressed syllables and higher energy to stressed vowel nuclei.

E2a/b & T2a/b (Heteronyms & Homonyms): E2 contrasts noun 'conduct' (E2a) with verb 'conduct' (E2b); standard dictionary G2P without syntactic POS disambiguation assigns identical phoneme sequences, resulting in highly similar F0 contours. T2 contrasts Telugu 'pattu' meaning grip (T2a) vs silk (T2b), where phrasal position induces subtle F0 and energy modulation.

E3 & T3 (Typographical Intra-Word Whitespace): Preserving the raw text in 'rhyth m' (E3) and 'ada vilo' (T3) inserts an explicit word-boundary pause token (SP) inside the split word. This is clearly visible in the Duration plots as an orange SP bar mid-word accompanied by a localized Energy drop, and causes isolated 'm' in E3 to be pronounced as 'EH1 M'.

E4 & T4 (Text Normalization & Code-Switching): Times ('10:45') and years ('2026') are expanded into spoken words in both languages, producing extended multi-token contours. In E4, the Romanized Telugu word 'andaroo' inside an English sentence is approximated via neural G2P fallback rules with English vowel quality.

5. Comparative Observations, Limitations & Reproducibility

Telugu's phonemic abugida script allows direct character-level modeling and faster convergence (Epoch 10) than English's opaque orthography, whereas Svarah's multi-speaker variety regularizes variance predictors better than studio IndicTTS. Future improvements include neural HiFi-GAN vocoder fine-tuning, POS-aware G2P for heteronyms, and bilingual phoneme sets for code-switching.